Figures 11.3 and 11.4. Physically, though, Plesser and I could find no associated calamity. Moreover, inspired by the observations of Aspinwall and Lütken (as well as a previous paper of theirs with Graham Ross), Plesser and I realized that we could repair the pinch mathematically in two different ways. One way led to the Calabi-Yau shape in Figure 11.3(a) while the other led to that in Figure 11.4(d). This suggested to us that the evolution from Figure 11.3(a) through Figure 11.4(d) was something that could actually occur in nature.

By late 1991, then, at least a few string theorists had a strong feeling that the fabric of space can tear. But no one had the technical facility to definitively establish or refute this striking possibility.

Inching Forward

Off and on during 1992, Plesser and I tried to show that the fabric of space can undergo space-tearing flop transitions. Our calculations yielded bits and pieces of supporting circumstantial evidence, but we could not find definitive proof. Sometime during the spring, Plesser visited the Institute for Advanced Study in Princeton to give a talk, and privately told Witten about our recent attempts to realize the mathematics of space-tearing flop transitions within the physics of string theory. After summarizing our ideas, Plesser waited for Witten's response. Witten turned from the blackboard and stared out of his office window. After a minute of silence, maybe two, he turned back to Plesser and told him that if our ideas worked out, "it would be spectacular." This rekindled our efforts. But after a while, with our progress stalled, each of us turned to working on other string theory projects.

Even so, I found myself mulling over the possibility of space-tearing flop transitions. As the months went by, I felt increasingly sure that they had to be part and parcel of string theory. The preliminary calculations Plesser and I had done, together with insightful discussions with David Morrison, a mathematician from Duke University, made it seem that this was the only conclusion that mirror symmetry naturally supported. In fact, during a visit to Duke, Morrison and I, together with some helpful observations from Sheldon Katz of Oklahoma State University, who was also visiting Duke at the time, outlined a strategy for proving that flop transitions can occur in string theory. But when we sat down to do the required calculations, we